

14 March 2025

GUIDELINES FOR DISTRIBUTORS AND TRANSPORTERS OF SEALED SOURCES

This guideline document deals with compiling and submission of sealed source monthly reports. It also deals with cradle-to-grave tracking process to account for all sealed radioactive sources that are used in South Africa. The tracking covers the manufacture, storage, use, transport, disposal (sale, transfer or discarding) of sources in South Africa, as well as import and export.

As part of this process, distributors and transporters of radioactive sources are required to submit reports and supporting documents monthly. The monthly report summarises all transactions involving radionuclides during the relevant month.

Document History

Final Version	Reason for Amendment	Effective Date
1	First issue and published for implementation	May 1998
2	Content structured on the new SAHPRA Guideline Template	January 2025

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Contents

Document History 1

Glossary 4

1. INTRODUCTION 4

1.1 Purpose 5

1.2 Scope 5

2. LEGAL PROVISION 5

3. Procedures for distributors and manufacturers..... 5

4. REFERENCES 8

5. VALIDITY 8

Glossary

Abbreviation/ Term	Meaning
NECSA	South African Nuclear Energy Corporation
MBq	megabecquerel, a unit of activity
KBq	kilobecquerel
GBq	gigabecquerel
TBq	terabecquerel
mSv	millisievert, a unit of equivalent dose (10 ⁻³ sievert)
POD	Proof of delivery
PDF	Portable document format
SAHPRA	South African Health Products Regulatory Authority
NLM	Nuclear liability management

1. INTRODUCTION

The regulation published under Government Notice R247 in Government Gazette 14596 of 26 February 1993 a sealed source is “a Group IV hazardous substance that is firmly bonded within solid inactive material or sealed in an inactive capsule of sufficient mechanical strength to exclude the possibility of contact with such substance and of the dispersion thereof into the environment under foreseeable conditions of use and wear; but this definition shall not apply where such bonding or encapsulation is solely for the purpose of storage, transport or disposal”.

A sealed source is a radioactive material (Group IV hazardous substance) that is either:

1. Permanently bonded within solid, non-reactive material, or
2. Encased in a durable, inactive capsule

This containment ensures:

- No direct contact with the radioactive material,
- No environmental contamination, and
- Safety during normal use and expected wear.

1.1 Purpose

These guidelines provide clear instructions for monthly report submissions under the relevant legislation. They outline the correct procedures and reporting requirements to help ensure accurate and compliant submissions from all users.

1.2 Scope

These guidelines apply to all distributors and transporters of sealed sources who supply end users of sealed sources. It covers sealed sources inventory management, movement tracking, disposal records, and usage. The requirements outlined apply to authority holders who distribute and transport sealed sources.

2. LEGAL PROVISION

- Hazardous Substances Act, 15 of 1973
- Regulations related to Group IV Hazardous Substances, R247 of 1993

3. PROCEDURES FOR DISTRIBUTORS AND MANUFACTURERS

Distributors and manufacturers should use the GLF-RDN-RN-14A form to account for transactions involving sealed sources on a monthly basis. The GLF-RDN-RN-14A form should be completed, signed and dated, and be emailed to: radionuclides@sahpra.org.za. If you are working electronically, you can add extra lines to the form as needed, then convert the completed form to PDF format, sign and submit it (Please do not omit columns or re-organise the form). It is helpful to us if you submit the electronic MS Word or MS Excel document along with the PDF version.

- All columns must be completed; however, a container serial number is not required where the source is packaged in an unnumbered lead pot or similar protective holder.
- Under Activity, use K, M, G or T to indicate KBq, MBq, GBq or TBq respectively.

The GLF-RDN-RN-14A form specifies the *supporting documentation that must be included* with the monthly report.

- *The source calibration certificates* must be copies of the manufacturer's original certificates, specifying the radionuclide involved, the serial number and the activity on a given date.

- Attach valid authorities of manufacturers and customers. This applies wherever the form asks for authority numbers.
- *Waybills or delivery notes* must be attached so that SAHPRA can find each listed source on its corresponding proof of delivery. The waybill/delivery notes must specify the radionuclides, source serial numbers and (where applicable) container serial numbers (A “container” here means any equipment that contains a source of radioactivity).

**Note that monthly returns submitted without the required supporting documents will not be processed.*

3.1 Sources brought forward from previous months

The figure you enter in Section 3.1 must correspond to the figure you entered in Section 3.5 of the previous month’s report, because the number of sources in stock at the beginning of any month will always be the same as the number in stock at the end of the previous month.

Write this figure in the shaded box marked *N1* (for number of sources in Section 3.1).

3.2 Sources acquired during this month

There are three ways in which distributors generally acquire sources:

- (a) purchase from a supplier, manufacturer or other local authority holder.
- (b) accept from an authority holder for delivery to NECSA for final disposal; or
- (c) import from another country.

Note that Section 3.2 is divided into three parts, corresponding to the above three ways of acquiring sources.

- (d) For purchase from local producers/suppliers, attach a delivery note and source certificate as described earlier, and check that you have a copy of each buyer’s current authority to possess and use the sources.
- (e) For sources accepted from a local authority holder (client) for final disposal (including exchange), attach the following documents:
 - The client’s letter W02, received from SAHPRA after submission of the GLF-RDN-RN-14A. This letter gives the authority holder permission to send the sources to NECSA for final disposal.
 - The client’s source list, with ticks indicating the sources to be disposed of.
 - A delivery note proving that the sources were collected from the client.
- (f) For imports, include source certificates, waybills, and, if possible, copies of each supplier’s authority or licence (from their regulator) to export the sources.

Document checklist for Section 3.2

	Purchase	From clients for disposal	Import
Source certificate			
Waybill/delivery note (proof of delivery)			
Client's exchange form OR Letter W02 from SAHPRA			
Client's source list with depleted sources			
Foreign supplier's authority (if possible)			

Add up the number of sources acquired and fill in the total in the shaded box marked N2 (for number of sources in Section 3.2).

Check before submitting: All sources listed in Section 3.2 must also appear as being “in stock” (Section 3.4) at the end of the month, unless they have already been disposed of, in which case they must be listed under sale, final disposal or export (Section 3.3.)

3.3 Sources disposed of during this month

There are three ways in which distributors generally dispose of sources:

- (a) sale to local authority holders;
- (b) delivery to NECSA for final disposal; or
- (c) export (for sale to a foreign buyer or return to a supplier/manufacturer).

Section 3.3 is divided into three parts corresponding to these three routes of disposal.

- (a) For **sales** to local authority holders, attach the delivery note and check that you have a copy of the buyer's current authority to possess/use the sources.
- (b) For final **disposals**, provide proof that NECSA received the specified sources:
 1. a copy of the waybill or delivery note detailing the sources; and
 2. confirmation of receipt by the Nuclear Liability Management (NLM) department at NECSA.

- (c) For **exports**, provide the waybill and, if possible, a copy of the recipient's authority/licence (from their country's regulator) to import and receive the sources. If an export authority was made out but no waybill number is given, the source must be indicated as being still "in stock".

Document checklist for 3.3

	Local sale	Final disposal	Export
Recipient's authority to possess & use/import			
Signed waybill/delivery note or proof of export, e.g. customs clearance certificate			
NECSA/NLM receipt on RN525 form			

**Sources sent to NECSA for disposal cannot be removed from the stock file before Radiation Control receives the signed receipt from NECSA.*

Add up the number of sources disposed of (sold, discarded and exported) and fill in the total in the shaded box marked N3 (for number of sources in Section 3.3).

Check before submitting: Any sources disposed of during the month (Section 3.4) will *not* appear in the "in stock" list (Section 3.4).

3.4 Sources in stock at month-end

Sources will be in stock if they were:

- 'Brought forward' from previous months (Section 3.1), or
- Acquired during the month in question (Section 3.2)

and were **not** disposed of (Section 3.3) during the month under consideration.

- Sources that are in transit to other authority holders will be counted as "in stock" until proof of delivery (POD) is provided.
- Similarly, sources that are on an export authority but have not yet passed through Customs will still be counted as being "in stock".

Count the sources in your stock list and fill in the number in the shaded box marked N4 (for number of sources in Section 3.4).

3.5 Check sum of month-end stock

This is a simple sum using the totals in the shaded boxes at the end of each section:

The number of sources brought forward (Section 3.1)			N1
PLUS	The number of sources acquired (Section 3.2) during the month	+	N2
MINUS	The number of sources sold, disposed of, or exported (Section 3.3)	–	N3
EQUALS	The number of sources in stock at the end of the month	=	

The answer must correspond with N4, the number of sources listed in Section 3.4.

Remember to enter your name and contact details and sign your form.

4. REFERENCES

Regulation published under Government Notice R247 in Government Gazette
14596 of 26 February 1993

5. VALIDITY

These guidelines are valid for a period of five (5) years from the effective date of revision and replace the Guideline for Distributors Sealed Sources form GLF-RDN-RN-14A. They will be reviewed in line with this timeframe or as and when required. Regulation published under Government Notice R247 in Government Gazette 14596 of 26 February 1993